

CrystalRL: Measuring and Engineering Legibility in Reinforcement Learning

Project proposal (write-the-paper-first draft)

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July 2026

Abstract

When an RL agent acts in a high-stakes domain, “why did it do that?” must have a verifiable answer — and if the agent is allowed to *modify itself*, legibility becomes a safety precondition, not a convenience. We propose a measurement-first research program on interpretable RL, built on a working three-generation system (a black-box hierarchical trader, its best post-hoc explanation program, and a successor rebuilt legible-by-construction) and one scalar, $\text{CRYSTALSCORE} = F \times S \times St$, on which all of them can be scored and *lose*. Preliminary results: rebuilding the core moved the score from 0.15 to 0.92 on real market data, the entire gap being simulatability. We propose three studies the existing infrastructure makes cheap and decisive: (A) does machine simulatability predict *human* simulatability (psychophysics); (B) a legibility frontier — how the interpretability–return tension depends on the substrate, mapped on designed markets with planted information value; (C) interpretability *drift* — whether names stay faithful while an LLM-driven loop modifies the agent, and whether the improver’s own rationales predict its effects. Each study has a pre-registered kill test. This draft is circulated before the studies are run, precisely to collect design criticism while it is still cheap.

1 Motivation and formal problem

Post-hoc explanation tells a story *about* a black box while the box goes on being one; Rudin [2019] argues high-stakes models should be interpretable by construction. What the field lacks is a measurement discipline that lets that thesis *fail*: most interpretability claims are not scored on an axis where a black box could win [Doshi-Velez and Kim, 2017, Lipton, 2018]. Our companion systems paper contributes the worked example; *this proposal is about the open questions it exposed*, which we believe are general.

Fix a policy π , a *story budget* K (at most K named rules/leaves), and a performance tolerance ε . Define

$$\text{CRYSTALSCORE}(\pi) = F \times S \times St, \quad (1)$$

$$F = \mathbb{E}[\mathbf{1}[\pi(b \leftarrow \text{do}(b^*)) = \pi^*(b^*)]] \quad (\text{faithfulness: do-writes land, judged vs a reference optimum}), \quad (2)$$

$$S = \text{acc}(\text{story}_K, \pi) \text{ s.t. } |\mathcal{R}(\text{story}_K) - \mathcal{R}(\pi)| \leq \varepsilon \quad (\text{simulatability: a } K\text{-rule story at performance parity}), \quad (3)$$

$$St = \frac{1}{n} \sum_i \mathbf{1}[\text{mechanism replicates on seed } i] \quad (\text{stability across training seeds}). \quad (4)$$

The parity constraint in Eq. 3 blocks the standard cheat (a story that explains a lobotomized policy); the do-operator in Eq. 2 [Pearl, 2009] blocks the other one (correlational “steering”). Three research questions:

- RQ1.** Is machine simulatability a valid proxy for *human* simulatability — the quantity that actually matters for oversight?
- RQ2.** What governs the return $\leftrightarrow$ legibility tension: the *substrate* (how much priced structure the environment has) or the *architecture*? Is there a scaling law linking optimal vocabulary size K^* to substrate complexity?

RQ3. Does legibility *survive self-modification*: do named coordinates drift as an automated improvement loop edits the agent, and are the improver’s stated rationales predictive of its measured effects?

2 Preliminary results: the instruments exist and discriminate

Three generations of one trading agent, evaluated on real market data (Table 1; details in the companion draft). **CHRL**: a constrained hierarchical PPO trader — three readable decision layers, five guardrails; competent (walk-forward Sharpe 1.09, 93% of benchmark Sharpe at 52% less drawdown), but its core is a 64-d unnamed latent with ~ 214 knobs. **Interpretable-CHRL**: the strongest post-hoc program we could run over the frozen policy — an eight-primitive discovered vocabulary whose targeted edits survive matched-random controls (+0.19 to +0.30 pp control-adjusted) — which also measured its own ceiling: a $K \leq 9$ story reproduces only 0.244 of deployed behavior, and no bound exists on what a latent edit touches. **CRYSTAL-1**: the core rebuilt so the ceiling disappears — memory is a named regime posterior $b_t = \mathbb{P}(\text{bear} \mid \mathcal{F}_t)$ (an explicit Bayes filter), the policy is a readable rule at parity, commands are five named levers. One certified rule (“ $b_t > 0.66 \Rightarrow \text{exposure } 0.74$ ”) held one-shot on 629 untouched trading days (Sharpe 1.46 vs 1.39, max drawdown -12.9% vs -15.5%).

	F	S (story, $K \leq 9$)	St
CHRL (black-box core, post-hoc program)	1.00	0.244	0.619
CRYSTAL-1 (legible by construction)	1.00	0.92	1.00

Table 1: The identical scalar on both architectures, real data. CRYSTALSCORE: 0.151 vs 0.92. Two methodological findings shape the proposal: (i) faithfulness does *not* discriminate — steering “works” on a black box too; the discriminating axis is simulatability; (ii) our first faithfulness probe silently read $F=0$ on a policy that *was* the rule by construction (fixed-size writes never crossed the decision boundary of a threshold policy) — interpretability metrics are instruments and need calibration surfaces.

Two further assets matter for the proposed work: a *designed-market* harness (synthetic regime environments where the value of belief information is planted by construction, fenced from all real-market claims by a value-of-information gate) and a running *LLM-driven improvement loop* whose frozen statistical gate has already certified real changes and refused fake ones (7 accepts, 0 placebo accepts on the certified rule’s history).

3 Proposed studies

Study A — does machine simulatability predict human simulatability? (RQ1) *Hypothesis*: machine S (Eq. 3) is a valid, cheap proxy for human prediction accuracy. *Design*: psychophysics on the real policy tables — participants see a K -rule story and predict the policy’s action in unseen states; conditions cross architecture (CHRL codes vs CRYSTAL-1 rules) $\times$ story budget ($K \in \{4, 9, 16\}$); the outcome is human accuracy vs the machine score on identical splits. No deception; think-aloud on a subsample for error taxonomy. *Why this way*: running user studies *first*, unanchored, is expensive and unfalsifiable — validating a machine proxy once lets every later iteration be measured for free [Doshi-Velez and Kim, 2017, Hoffman et al., 2018]. *Kill test*: human accuracy at chance while machine S is at ceiling kills the proxy (and with it, our headline 0.92 becomes a machine-only claim — worth knowing). *Status*: *not yet run; no human data collected*.

Study B — the legibility frontier (RQ2). *Hypothesis*: the return $\leftrightarrow$ legibility tension is a *substrate* property: it vanishes where belief information carries value and binds where the regime

is fully priced. *Design*: a family of designed markets with the planted value of belief information swept from 0 to large (our harness already produces the endpoints: value +282% of a belief-blind baseline on one end; a certified return–legibility tension on a hard substrate on the other); at each point, train matched pairs (free policy vs budget- K readable policy for several K) and record the frontier (substrate complexity, return gap, CRYSTALSCORE); then locate the two real panels we operate on that map. The output is a *phase diagram*: where legibility is free, where it costs, and how K^* scales. *Why this way*: on real data alone a null is uninterpretable (the instrument and the substrate are confounded — our value-of-information gate exists precisely because a perfect belief can be worth zero on a priced market); designed substrates give ground truth, and the gate keeps their conclusions quarantined from market claims. *Kill test*: tension persisting on a high-planted-value substrate kills the substrate hypothesis (the tension would be architectural); optimal K^* unrelated to measured complexity kills the scaling law. *Status*: harness built; the sweep and the phase diagram do not exist yet.

Study C — interpretability drift under self-modification (RQ3). *Hypothesis*: under an automated improvement loop, naming faithfulness decays unless actively re-certified — “bear” gradually stops meaning bear — and the improver’s stated rationales are only weakly predictive of its measured effects. *Design*: run the existing loop for ≥ 20 accepted modifications on the real machinery; after each accept, re-measure Eq. 2 and the naming audit (is the state called “bear” still the state where de-risking is optimal under the world model?); separately, score every proposal’s written rationale against its measured effect (sign agreement, calibration). *Why this way*: this is the one study our infrastructure makes uniquely cheap — the loop, the frozen gate, and the audit all exist and have operating history; and it directly tests the safety claim that motivates the whole program (*a self-modifying agent must remain readable while it changes*). *Kill tests*: F staying at 1.0 over 20+ accepted changes with no re-certification kills the drift hypothesis (good news — legibility is self-maintaining); rationale–effect agreement at chance is itself a publishable confabulation result [Jacovi and Goldberg, 2020]. *Status*: the loop exists; the longitudinal drift protocol has not been run.

4 Why this program and not the alternatives

Why not post-hoc explanation at scale (saliency, probing, concept extraction)? We ran the strongest version we could build: it found real, editable structure (that is honestly reported), but it cannot bound an edit’s blast radius, name a command, or compress the policy into a story — measured, not asserted. *Why not user studies first?* Without a validated machine proxy, every design iteration costs a new human experiment; Study A is sequenced first precisely to unlock cheap iteration. *Why not purely synthetic benchmarks?* A synthetic result does not transfer by default; our value-of-information gate (designed-market conclusions are quarantined until the same measurement is positive on a real book) is the explicit bridge discipline, and Study B uses synthetic substrates only where real ones are structurally uninformative. *Why a product score rather than a dashboard?* A single scalar with a fixed budget can *lose* — and its known failure modes (probe blindness on threshold policies; faithfulness non-discrimination) are documented and repaired rather than hidden.

5 Gaps this draft does not close

(i) All preliminary evidence is from *one* domain (daily-frequency trading) and largely one substrate per claim; the studies are designed to widen this, but domain transfer beyond finance (e.g., robotics, dialogue agents) is future work. (ii) S is currently measured against policy *tables*; extending parity-constrained simulatability to stochastic policies needs a definition choice (we propose total-variation-bounded story matching — feedback welcome). (iii) The stability axis

uses seed replication as a proxy for mechanism identity; a stronger identifiability criterion would be better and we do not have one. (iv) Study A’s power analysis depends on between-subject variance we cannot know in advance; we will pilot with $n=8$ before committing the full design. (v) The drift study’s negative control (a loop with re-certification disabled) doubles compute; if resources bind, we run it on the designed substrate only.

6 Plan, resources, and the ask

Timeline (6 months): M1 — Study A pilot ($n=8$) + power analysis; freeze the psychophysics protocol. M2–M3 — Study A full run; Study B substrate sweep (compute-bound, parallelizable). M4 — Study B phase diagram + real-panel placement. M5–M6 — Study C longitudinal run (≥ 20 accepts) + rationale-calibration analysis; write-up. All three studies are pre-registered with the kill tests above before data collection; every experiment logs to the project’s public logbook under its honesty contract (every entry names its null, its seed, and its worst caveat).

Resources: the codebase, data panels, designed-market harness, and improvement loop are built and operating; Studies B–C are CPU-scale. The one new requirement is Study A’s participant pool (target $n \approx 30$; standard ethics review for anonymous behavioral-prediction tasks).

The ask: this draft is circulated *before* any of the three studies has been run, in the write-the-paper-first spirit [Eisner, 2010]: criticism of the designs — especially Study A’s conditions and the Eq. 3 extension of gap (ii) — is worth far more now than after the data arrive. If the program interests you, any of the three studies stands alone as a supervised project; Study A is the natural first collaboration.

References

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